

NumPy & Pandas

Load, clean and explore data · Equigrow short notes

NumPy

```
import numpy as np
a = np.array([1, 2, 3, 4])
a * 2          # [2 4 6 8] (no loop needed)
a.mean(), a.max(), a.reshape(2, 2)
m = np.zeros((3, 3)); m[0, 1] = 5
```

Pandas: load & look

```
import pandas as pd
df = pd.read_csv("students.csv")
df.head(); df.info(); df.describe()
df["cgpa"]          # one column
df[df["cgpa"] > 8]   # filter rows
df.sort_values("cgpa", ascending=False)
```

Clean

```
df.isna().sum()          # missing values per column
df["age"] = df["age"].fillna(df["age"].median())
df = df.drop_duplicates()
df["branch"] = df["branch"].str.strip().str.upper()
```

Group, merge, plot

```
df.groupby("branch")["cgpa"].mean()
merged = pd.merge(students, marks, on="id", how="left")
df["cgpa"].plot(kind="hist")
```